D Additional comparisons between Pre-Norm and QKV-Norm

D.1 Model performance

Table 8 shows the per-token accuracy performance for the trained `Baseline` models. `Pre-Norm` and `QKV-Norm` have comparable in-distribution per-token accuracies of 91.4% and 91.0% respectively. However, performance drops to 87.5% (82.5%) for generalisation to intermediate task difficulty, and 66.7% (46.8%) for increased difficulty. The performance drop of `QKV-Norm` implies that it has learned a less generalisable solution. This re-enforces our motivation that architectural changes should be important for the inductive bias of a model.

Dataset	`Pre-Norm`	`QKV-Norm`
In-distribution	$91.38 \pm 0.04\%$	$90.99 \pm 0.03\%$
OOD (interpolation)	$87.46 \pm 0.04\%$	$82.54 \pm 0.04\%$
OOD (extrapolation)	$66.65 \pm 0.05\%$	$46.76 \pm 0.05\%$

Table 8: Per-token accuracy for the `Baseline` models. Dataset configurations are shown in Table 3.

D.2 Training stability

Changing the normalisation layer is expected to affect the training rate and stability. To investigate this, Figure 24 shows the training curves for different model sizes and learning rates. The task is configured as presented in Table 7. The `Depth` parameter is the number of layers, where brackets indicate the values for an encoder-decoder model. For example, $(2, 2)$ means that we use 2 encoder blocks and 2 decoder blocks. Each decoder block has a self-attention and a cross-attention layer, and so the total model has 6 attention layers. A single `Depth` value indicates a decoder architecture, with the number of layers shown. `Width` is the number of neurons per layer, and `Latent width` is the number of neurons on the space of $\{q,\ k_t, v_t\}$ (called N_{qkv} in section 4). Training curves on the top row use a learning rate of 0.001, whilst the bottom row use a value of 0.0001. In each panel, two training runs are shown, with different random seeds. `Pre-Norm` is shown in blue, and `QKV-Norm` in red.

We find that `Pre-Norm` training is unstable for large learning rates and model sizes, as shown by the flat blue curves in the top right hand panels. Similar stabilisation improvements at large learning rate is reported for `QK-Norm` in [19, 20], which applies layer normalisation to $\{q,\ k_t\}$ but not v_t, as for `QKV-Norm`. However, we note that training large models with a smaller learning rate leads to improved model performance, as shown by the panels on the bottom right. Finally, we note that both methods typically train the model at similar rates, however small model training follows a very different trajectory, with `QKV-Norm` learning more slowly at the beginning of training (bottom left panels). There is also some visible evidence that small model training is actually *less* effective when using `QKV-Norm` with a large learning rate (top left panels).

D.3 Attention sparsity

We find that our `Pre-Norm` models often exploit sparse-attention, whereas models trained with `QKV-Norm` do not. Similar behaviour is reported for `QK-Norm` in [18]. For a systematic comparison, Figure 25 shows a histogram of the maximum attention observed per-distribution (i.e. a histogram of $\max_i a_i$). When making this plot, we do not consider the first row of the attention matrix, in which the `[` token attends fully to itself.

We see that the `Pre-Norm` distribution has a sharp peak at 1, indicating a significant use of sparse-attention. By contrast, the `QKV-Norm` distribution is weighted towards 0 and has no peak at 1. To verify this behaviour, Figure 26 shows an attention heatmap for a randomly chosen datapoint when using the `Baseline` `Pre-Norm` model, and Figure 27 shows the same datapoint for `QKV-Norm`. We observe a significantly less sparse attention matrix for `QKV-Norm`. Note that [18] also shows a similar visualisation.

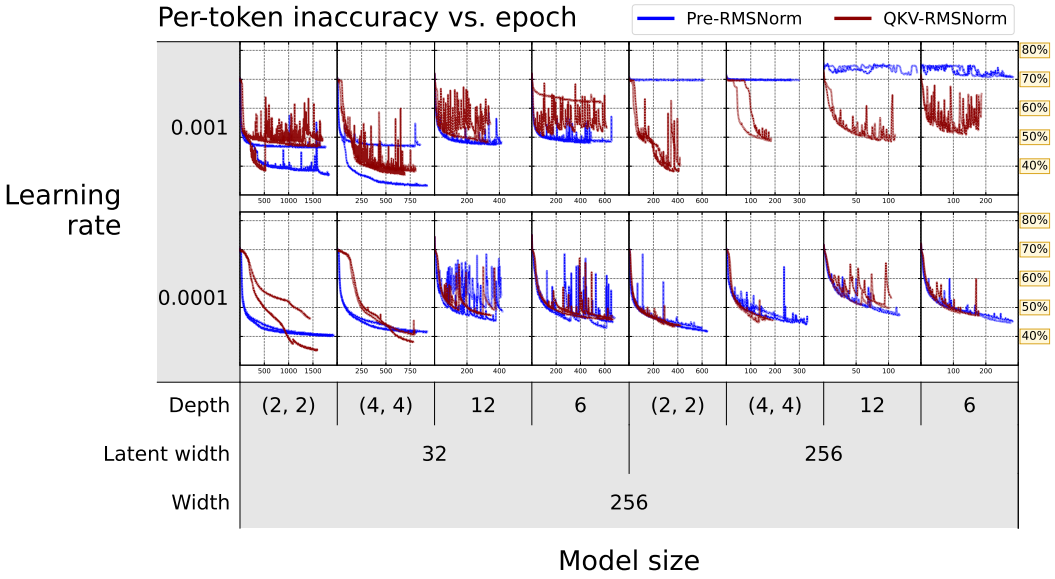

Figure 24: Training curves when learning the task configuration shown in Table 7.

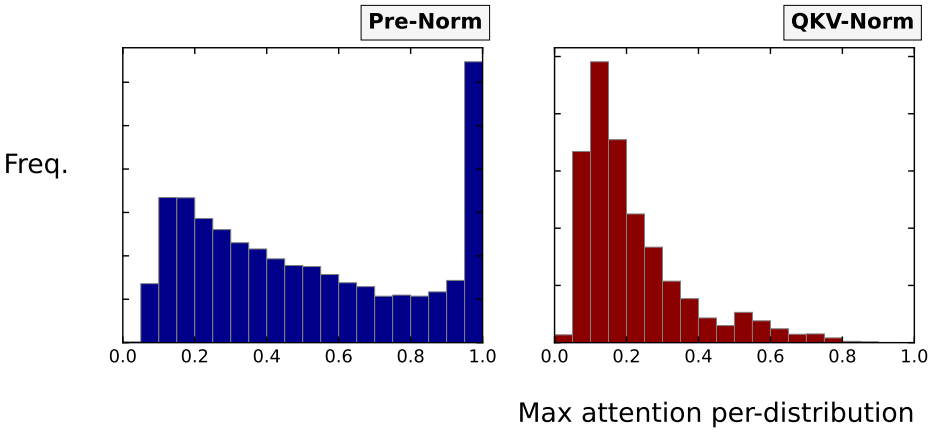

Figure 25: Distribution of the maximum attention observed per-distribution, i.e. $\max_i a_i$, in the `Baseline` case. We observe that the `Pre-Norm` model often utilises sparse-attention, as seen by the peak at 1. By contrast, `QKV-Norm` shows no such peak. Similar behaviour is reported for `QK-Norm` in [18].

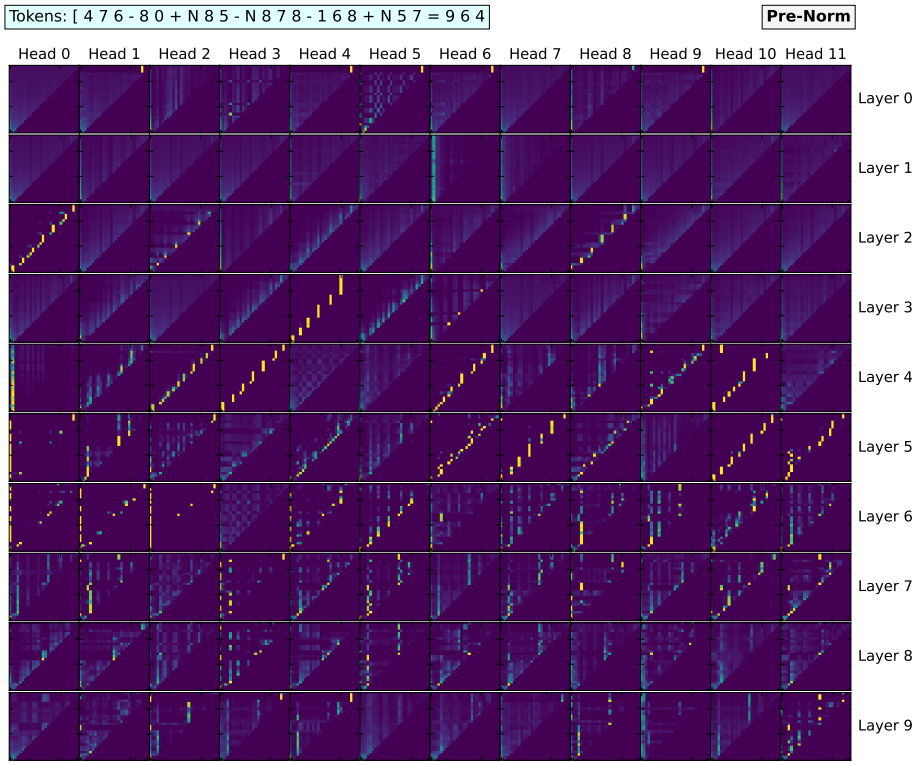

Figure 26: Attention maps for a random in-distribution example using the `Baseline Pre-Norm` model. Several attention heads create sparse attention distributions.

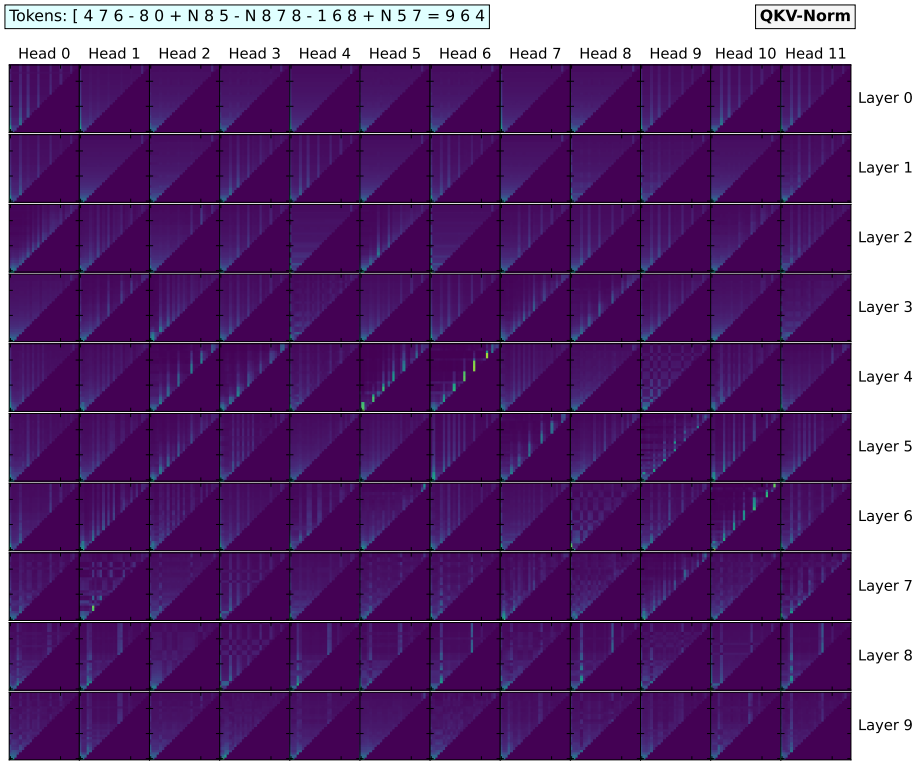

Figure 27: Attention maps for a random in-distribution example using the `Baseline QKV-Norm` model. We observe much less sparsity than in the `Pre-Norm` model, shown in Figure 26. Similar behaviour is reported for `QK-Norm` in [18].